MACHINE LEARNING • LINEAR MODELS

ELASTIC NET REGRESSION

A hybrid regularized linear model combining L1 and L2 penalties to achieve optimal feature selection and stability.

01 • ELASTIC NET OBJECTIVE

How does Elastic Net mix regularization penalties?

The loss function uses a weighted mixture of Lasso (L1) and Ridge (L2).

$$\text{Loss} = \text{SSE} + \lambda \times [r \times \Sigma|w_i| + (1 - r) \times \Sigma w_i^2]$$

SCIKIT-LEARN CONVENTION

`alpha` controls overall strength (λ).

`l1_ratio` controls the mixture (r).

SSE (SUM OF SQUARED ERRORS)

Base prediction error on training data.

Λ (LAMBDA)

Overall regularization strength multiplier.

R (MIXING PARAMETER)

Controls the L1 vs L2 ratio (between 0 and 1).

$\Sigma|w_i|$ (L1)

Lasso absolute penalty.

Σw_i^2 (L2)

Ridge squared penalty.

* As always, the intercept (w_0) is generally unpenalized.

Combining shrinkage and selection to stabilize models.

THE BEST OF BOTH WORLDS

L1 Regularization: Forces weak coefficients exactly to zero, performing automatic feature selection.

L2 Regularization: Smoothly shrinks coefficients toward zero without discarding them entirely.

Synergy: Selects features while stabilizing highly correlated groups.

UNDERFITTING RISK

Applying strong regularization across two distinct penalties can rapidly strip predictive power from the model if λ is set too high.

THE MIXING PARAMETER (r)

r = 0 (Pure Ridge)
100% L2 penalty. Shrinkage only, no exact zeros.

r = 0.5 (True Elastic Net)
Balanced mixture of L1 and L2 penalties.

r = 1 (Pure Lasso)
100% L1 penalty. Maximum sparsity and selection.

03 • NUMERICAL EXAMPLE

Calculating the combined Elastic Net objective function.

INITIAL MODEL STATE

$$\hat{y} = 10 + 100x_1 + 80x_2$$

GIVEN PARAMETERS

SSE = 500

$\lambda = 0.1$

$r = 0.5$

$w_1 = 100, w_2 = 80$

1. L1 TERM (R=0.5)

$L1 = 0.5 \times (|100| + |80|)$

L1 = 90

2. L2 TERM (1-R=0.5)

$L2 = 0.5 \times (100^2 + 80^2)$

L2 = 8,200

3. COMBINED PENALTY & TOTAL LOSS

$Penalty = \lambda \times (L1 + L2) = 0.1 \times (90 + 8,200) = 829$

Loss = 500 + 829 = 1,329

** Note: This is the objective value for the given starting coefficients. The model will learn updated coefficients by minimizing this value via gradient descent (controlled by learning rate η).*

04 • TRADEOFFS

Evaluating the strengths and limitations of Elastic Net.

ADVANTAGES

Best of Both Regularizations
Combines Ridge's stability with Lasso's sparsity.

Correlated Feature Handling
Unlike Lasso, which drops correlated features randomly, Elastic Net shrinks them together as a group.

Overcomes "p > n" Limit
Lasso can select at most 'n' features. Elastic Net bypasses this limitation, useful for vast datasets.

DISADVANTAGES

Complex Hyperparameter Tuning
Requires cross-validation across a grid of both regularization strength (λ) and mix ratio (r).

Requires Feature Scaling
Data must be strictly standardized for the dual penalties to apply fairly.

Complex Interpretation
Coefficient paths are harder to trace than pure L1 or L2 models.

Summary of linear regularization techniques.

ASPECT	RIDGE	LASSO	ELASTIC NET
Regularization	L2	L1	**L1 + L2**
Coefficients	Shrinks toward zero	Can become zero	Can become zero
Feature Selection	No automatic removal	Yes	Yes
Correlated Features	Handles via shrinkage	May select one unpredictably	Often handles them more effectively
Main Use	Reduce coefficient magnitude	Feature selection	Feature selection with coefficient shrinkage

Conclusion: Elastic Net is the most robust generalized linear model when both feature selection and handling grouped correlated variables are critical to the project's success.